

AI-POWERED RETAIL INTELLIGENCE DIGITAL TWIN

A Deep Learning-Based Decision Support System for Small and Medium Retail Businesses

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ABSTRACT

Small and medium retail businesses often rely on manual decision-making for inventory management, demand forecasting, customer analysis, and promotional planning, which frequently results in stock shortages, overstocking, inefficient marketing, and reduced customer retention. Existing retail management systems primarily provide historical reports and basic analytics but lack intelligent predictive capabilities and simulation tools that help managers make proactive business decisions. This project proposes an AI-Powered Retail Intelligence Digital Twin, an intelligent decision-support platform that creates a virtual representation of a retail store using historical and real-time business data. The system integrates Machine Learning, Deep Learning, Natural Language Processing, Generative AI, and simulation to forecast sales, optimize inventory, analyze customer behaviour, predict footfall, assess review sentiment, recommend products, and generate business insights, with a "What-If Simulation" Digital Twin at its core allowing managers to test decisions virtually before implementing them in the real world.

The platform is built around the following integrated modules:

- **Sales Forecasting** — predicts daily and weekly sales and product demand using the Transformer-based PatchTST time-series model.
- **Inventory Optimization** — predicts remaining stock, restock quantity, and restock date using XGBoost.
- **Customer Segmentation** — groups customers into Premium, Loyal, New, and At-risk segments using K-Means clustering.
- **Product Recommendation** — suggests similar and frequently-bought-together products via Collaborative Filtering and Market Basket Analysis (Apriori/FP-Growth).
- **Customer Churn Prediction** — identifies customers likely to stop purchasing using XGBoost.
- **Customer Footfall Prediction** — forecasts daily visitors, weekend traffic, and festival demand.
- **Review Sentiment Analysis** — classifies customer reviews as positive, neutral, or negative using BERT, and surfaces recurring complaints.
- **AI Promotion Recommendation** — suggests promotions based on overstock, low demand, customer segment, and seasonal trends.
- **AI Business Copilot** — an LLM-powered assistant (Llama 3 / Gemma) that answers natural-language business queries such as restocking needs or churn risks.
- **AI Report Generator** — automatically compiles daily, weekly, and monthly business reports.

- **Digital Twin (core innovation)** — a virtual retail store that simulates festivals, discounts, traffic surges, inventory delays, and seasonal demand, predicting their effect on revenue, inventory, footfall, and promotion effectiveness.

The system is trained and validated using publicly available datasets, including Rossmann Store Sales, Olist Brazilian E-Commerce, UCI Online Retail II, and Instacart Market Basket Analysis, providing realistic data for forecasting, customer analytics, reviews, and recommendations without dependence on proprietary retail data. By unifying Deep Learning forecasting, Machine Learning analytics, NLP, recommendation systems, Generative AI, and Digital Twin simulation within a single decision-support platform, the project moves beyond conventional retail dashboards to an intelligent, predictive, and simulation-driven ecosystem, demonstrating strong technical depth and direct relevance to current trends in intelligent retail systems.

Keywords: *Digital Twin, Deep Learning, PatchTST, BERT Sentiment Analysis, Sales Forecasting, Customer Segmentation, Recommendation Systems, Generative AI, Business Simulation, Retail Intelligence*